



An Apple Counting System Robust to Multiple Intermittent Occlusions

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Abstract. Automated fruit counting, through the use of computer vision methods, is a very important step towards the development of precision agriculture, and, as a consequence, has been object of study by multiple authors. In this work, we present a novel method to detect, track and count fruits that is scalable and robust towards problems existing in other fruit counting methods. Namely, our approach consists in a method that uses (1) a stereo camera that can be used reliably under direct sunlight, (2) a fast detection algorithm, and (3) an algorithmic approach to track fruits which is robust to fruit occlusions. We applied our proposed solution in an apple orchard and were able to provide apple counts with an error ranging from 15% to 57%. The characteristics of this novel approach and the preliminary results achieved seem promising in order to tackle the problem of fruit counting with occlusions at the large scale of a whole orchard, since it addresses the problem of intermittent occlusions that is overlooked by other approaches. Even with errors, the overview of the entire orchard, resulting from the scalability of the process, which can be implemented with a camera in a tractor, makes real-time mapping possible, supporting technical decisions with major economical impact, such as supplementary fruit thinning, water or nutrient adjustments or harvest management.

Keywords: Apple Orchard · Fruit counting · Fruit tracking · Stereo vision

1 Introduction

1.1 Problem Description and Motivation

Recent improvements in sensor technology and the development of new computer vision techniques have made possible remarkable advancements towards

automated agriculture, from decision support systems for agronomic management, to the ambitious goal of automation of cultural practices by machines. In particular, there is a strong interest in automatically counting fruits in an orchard, a task that is currently being handled by manual counting estimations, which is time consuming, impractical on a large scale, and highly error-prone.

To address this problem, it is important to study, not just the algorithmic techniques to process data, but the sensors to collect it, as the feasibility of any proposed solution depends on the possible drawbacks of the tools available. In particular, we study the case of an apple orchard, although our system is general enough to potentially handle other kinds of fruits. This task has specific hurdles to be surpassed: the variability of fruit colour, size, orientation, visibility, and lightning conditions make the problem of detecting fruits challenging. Moreover, the intermittent occlusion of fruits by other fruits, branches and leaves is a major challenge to the problem of tracking and counting fruits.

We approach the problem in this work using a very simple setup composed by a single stereo camera, which is more affordable than most Digital Single-Lens Reflex (DSLR) cameras used in many similar works and performs better under sunlight conditions than most depth cameras studied. On the algorithmic part, we present a multi-step solution, beginning by detecting, then tracking, and finally counting the total number of fruits. On the detection end, we use a Deep Learning model for fast-detection. Regarding the tracking and counting part, we present a novel approach to fruit tracking based on 3D centroids which is robust to intermittent occlusions that happen when a fruit is being tracked, becomes occluded during a few frames, and reappears again later, and is computationally less demanding than a previously proposed approach to the same problem [8].

Our solution may be summarised as follows. First, an object detector, trained on images of fruits that we collected and manually annotated, is used to detect fruits in each video frame. Then, fruits are tracked in 3D across the video frames by an algorithm that we developed, in order to associate detections of the same fruit across the video frames and count each fruit only once. Finally, the total number of fruits can be recovered by simply counting the number of objects that the algorithm tracked in step 2, after applying some filters.

1.2 Related Work

Gongal et al. made a review [3] about the sensors, features and methods more commonly used for fruit detection and location and summarised the main issues that arise when tackling these problems, namely: variable lighting conditions, complex canopy structures, varying colour, size and shape of fruits, occlusions of fruits, clustering of fruits, and displacement of fruits by wind.

Several authors already tackled the problem of identifying and counting fruit in orchards, with a variety of methods. Gené-Mola et al. [2] used RGB images collected with a DSLR camera and processed through a Structure-from-Motion (SfM) [15] pipeline in order to detect and locate fruits in 3D. They applied the Mask R-CNN deep learning model to segment apples, while using

other faster detection methods such as the YOLO [11] would improve the performance of the system. Moreover, the high computational time needed to process the images with SfM is one limitation that has to be taken into account. Gongal et al. [4] used a combination of RGB images and depth maps collected with a time-of-flight depth camera to locate and count apples on an orchard. Wang et al. [16] used a stereo camera rig composed of two DSLR cameras to detect and count apples, by triangulating their 3D positions in the image sequence, and then projecting them onto a global coordinate system. The biggest limitation of both works is their fruit detection being heavily tied to colour features, which are very sensitive to variations in light and colour. Despite the good results in the identification of red apples, the methods struggle to identify apples in early phenological stages, when they are green and smaller in size. The solutions proposed by Nguyen et al. [9] and Tao et al. [14] used a low-cost RGB-D camera to photograph orchards and generate a 3D point cloud, and for fruit detection, they filter the point cloud by colour and apply clustering techniques. These solutions using depth cameras rely on infrared light and thus have limitations in outdoor conditions under direct sunlight, depending on a light shield, tunnel or some other device to block it and on the usage of LEDs to create the ideal lighting conditions. This adds complexity to the data collection process, and makes the solution less scalable if one intends to apply it in large scale to a whole orchard, for example, and to various conditions such as terrain slope. Stein et al. [13] and also Liu et al. [7] tackled the problem of counting and locating fruits with a monocular RGB camera by training a Faster R-CNN model. The biggest limitation of both approaches is that the tracking is done sequentially, with each detection being matched against the ones in the next image only. This means that a single false negative, i.e. a fruit that is not detected in a given image, inevitably breaks the tracking of that fruit, and a new track is introduced later on when the fruit is detected again in a subsequent image, leading the system to count two fruits instead of one.

To summarise, most of the works relying on depth cameras based on infrared light have limited applicability under direct sunlight, resorting to solutions such as light tunnels to block it. On the other hand, solutions relying on monocular RGB cameras tend to use SfM to produce 3D data and are thus computationally more demanding than stereo approaches. On the detection side, approaches relying on colour and geometric features to detect fruits through more traditional computer vision and/or clustering techniques are strongly affected by variations in fruit colour and light conditions, whereas approaches based on Deep Learning models are usually more robust. Therefore, we propose a solution to the fruit counting problem that is scalable – i.e. suitable to be adopted at a whole orchard level by a farmer –, robust to light and colour changes in outdoor conditions. We ruled out solutions relying on infrared light and light shields, and fruit detectors based solely on colour, and decided to go for a stereo solution in order to recover depth maps in real time.

2 Materials and Methods

2.1 Description of the Data

The data for the following tasks was collected in an orchard owned by INIAV [5] in Alcobaça, Portugal (Lat. 39,549 N; Long. -8,957 W), with a stereo camera—the ZED 2i—being held by hand, attached to a computer, by shooting video in roughly straight lines parallel to the line of trees. No stabiliser or any other type of helping device was used. Several lines of trees were captured, corresponding to different apple tree clones from the group *Gala*, when the fruits were close to harvest, corresponding to the phenological stages from BBCH 81 (beginning of ripening) to BBCH 87 (near fruit ripe for picking), according to BBCH-scale for pome fruits.

2.2 Detecting Fruits in 2D Images

Data Annotation. Our image annotations consist of bounding boxes around all the apples visible in each image. Video frames corresponding to the left view of the stereoscopic camera were extracted from the ZED proprietary video files and stored as image files. They were then manually annotated using the open-source tool *Makesense* [12]. Only one in every 30 frames was annotated due to the similarity between consecutive frames. We annotated a total of 183 images, each one containing approximately 100 fruits, on average, converging the aforementioned phenological states.

Model Training. For this work, a *Darknet* implementation of the YOLOv4 [1] was used. Darknet is an open source neural network framework written in C and CUDA [10]. The YOLOv4 is an object detection Deep Learning model belonging to the YOLO family, which was specifically designed to handle real-time object detection [11]. Unlike many of the deep learning models used in some of the previous works, this type of model is more suitable for in-field operations due to its speed. Although more recent YOLO models are available, we stick to v4 due to its license allowing commercial usage. Our YOLO model was first pre-trained on the MS COCO [6] dataset—a large scale image dataset containing 330K images over 80 different object categories. Then, we fine-tuned it to our own tasks, using our annotated dataset.

2.3 Tracking and Counting Fruits in 3D

Overview. Having a YOLO detector capable of detecting the fruits of interest in each video frame is not sufficient to count them. In fact, the same fruit will most certainly appear in several video frames, thus simply summing the number of detected boxes ends up with duplicate counts. We must track each fruit across frames in order to understand which detection boxes (in different frames) correspond to the same fruit, and then count only one of them. This is where 3D data becomes useful.

By using a Structure-from-Motion (SfM) algorithm, it is possible to assemble, from a set of 2D images, a 3D representation of the photographed scene in the form of a point cloud and/or a depth map. Another way of obtaining a point cloud is by using a depth camera or a stereo camera, such as the ZED 2i. These devices are designed to output a depth map – from which a point cloud can be directly generated –, thus avoiding the need to run computationally expensive SfM algorithms.

Having a dense 3D representation of the scene in the form of a point cloud, there is a matching between each pixel in a 2D image and a 3D point in space. This allows us to project each fruit detected in a 2D frame onto its 3D representation in a global coordinate system that is common to all frames. Therefore, whenever the same fruit is detected in different frames, the projection of the corresponding detection boxes will lead to the same region in 3D space, and so we can conclude that it is in fact the same object and thus count it only once.

Algorithm Description. In order to track and count fruits along a video, we implemented the algorithm proposed in [8] which works by reconstructing a 3D scene from the video and localising the fruits in 3D space. While the authors in [8] used a SfM pipeline to obtain the camera extrinsics and dense depth maps for the scene, which is a computationally demanding task, we used the ZED SDK instead to retrieve this data in real time from the ZED 2i stereo camera. As a result, we avoid several hours of computation and obtain a system that is less demanding in terms of computer resources and suitable to be setup and used in field, either manually operated or mounted on an agricultural vehicle. Moreover, we also implemented an algorithm similar to [8] but, instead of re-projecting the whole set of 3D points, it re-projects only the 3D centroid of each fruit. This algorithm is summarised in the block diagram shown in Fig. 1, and detailed in the pseudo-code of Algorithm 1.

For each video frame, we first run YOLO on that frame to detect fruits. Then, for each detected fruit, the pixels contained in its bounding box are projected onto the 3D global coordinate system using the frame’s depth map. Since some of these points belong to the fruit, but others do not – usually some background pixels are contained in the bounding box –, we compute the *median* of the coordinates of these points to obtain a *centroid* representing the fruit. The centroid is used to describe each encountered fruit throughout the video.

The centroids of all the fruits detected in the previous frames are reprojected into the 2D image plane of the current video frame by using Eq. 1:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \cdot [R|t] \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (1)$$

where (X, Y, Z) is a 3D point in the global coordinate system, R and t are the rotation matrix and translation vector, respectively, from world to camera coordinates, K is the camera intrinsics matrix, λ is the scale factor and (u, v) are

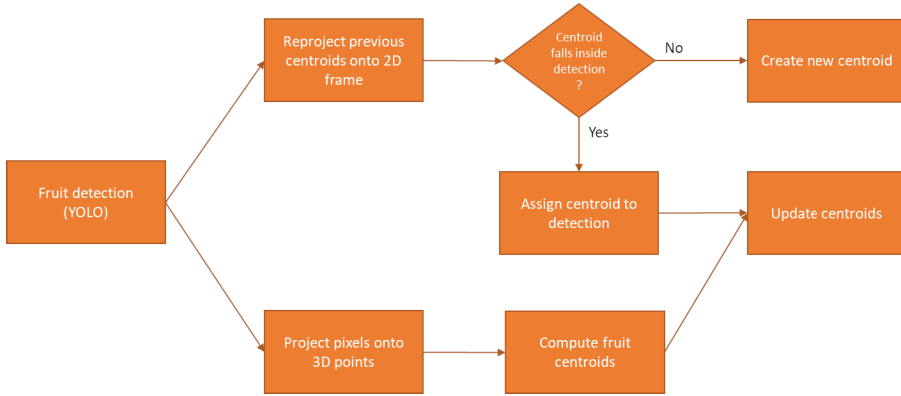


Fig. 1. Block diagram of the tracking algorithm steps for one frame. This procedure is repeated for every frame in the video. In the end, the number of fruits is given by the number of centroids.

Algorithm 1. Fruit tracking and counting algorithm

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1: procedure COUNT_FRUITS(video)
2:    $centroids \leftarrow \emptyset$ 
3:   for  $i = 1$  to  $num\_frames(video)$  do
4:      $frame \leftarrow next\_frame(video)$ 
5:      $detections \leftarrow YOLO\_detect(frame)$ 
6:      $reprojected \leftarrow reproject(centroids, frame)$  ▷ Equation 1
7:     for each  $detection$  in  $detections$  do
8:        $new\_centroid \leftarrow compute\_centroid(detection)$ 
9:       for each  $centroid$  in  $reprojected$  do
10:        if  $centroid \in detection.box$  then
11:           $centroid \leftarrow update(centroid, new\_centroid)$ 
12:           $detection.ID \leftarrow centroid.ID$ 
13:        else
14:           $detection.ID \leftarrow new\_centroid.ID$ 
15:           $centroids \leftarrow centroids + \{new\_centroid\}$ 
16:   return  $card(centroids)$ 

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the coordinates of the corresponding 2D projection into the frame. The camera intrinsics matrix K includes parameters such as the lens' focal length and the optical center, and is obtained by a calibration procedure. The scale factor λ appears in the equation due to the ambiguity induced by the loss of one dimension in the projection from the 3D world to the 2D image plane. We may simply divide the first two coordinates of the 2D projection by the third coordinate to convert homogeneous coordinates from projective space into euclidean space and obtain (u, v) .

If the 2D projection of a given centroid falls within a bounding box in the current frame, then the fruit in this bounding box is assumed to be the same

of that centroid, which means we are tracking a fruit that has been seen before, and so we assign it the same ID. If, however, there is no 2D projection falling inside a given bounding box, then we assume it corresponds to a new fruit and a new ID is assigned to it. We follow this procedure until the last video frame. In the end, the number of unique IDs is, in theory, the total number of fruits, without repetition.

Due to the error associated with the computation of the camera poses and the fruit centroids, the centroids sometimes lie closer to the surface of the fruit than to its center of mass. Therefore, when the camera sees the same fruit in a different angle, in a subsequent frame, the centroid projection may end up outside of the bounding box of that fruit by a small margin, originating a new detection with a brand new ID, making the algorithm wrongly count this fruit twice. To minimise this problem and improve the tracking stability, we update the coordinates of each centroid with each new frame where that same fruit is detected, by averaging over the centroids previously calculated for that fruit.

Another obstacle to this method is created by YOLO, which sometimes produces false positive detections. In order to mitigate these, in the end, when we are counting the unique IDs of the detected fruits, we ignore fruits that were detected in less than 5 frames in the whole video, as proposed in [8].

The algorithm that we propose here has the potential to be particularly suitable for real-time detection since, in each step, it only needs the current video frame, its depth map and the previously calculated centroid coordinates, instead of having to handle all the 3D points, like in [8]. The depth camera used is capable of calculating the depth map in real-time, and the rest of the computational resources are mostly spent on the 2D detection. Thus, given our practical choice of the YOLO model, the combination of our setup, algorithm, and model is ideal for in-field fruit detection.

3 Experimental Results

3.1 Test Procedure

We prepared three different scenarios to match as closely as possible the one in which our system is intended to be used, each corresponding to a different clone of the most important apple variety planted all over the world (*Schninga Schnico*, *Schnico Red*, *Galafab*). In the same orchard used for training the models (Sect. 2.1), we chose 3 portions of tree lines that were not used during training. For each portion, we chose 3 to 6 representative trees to evaluate our algorithm on, which were filmed on one side of the line only, and whose fruits were manually counted in the field and on the video. The selected trees are not necessarily consecutive, so the video frames were manually edited to mask all other trees before we passed them to the algorithm.

Fruit counts manually obtained in field are not a suitable ground truth to directly evaluate our system's performance, because the number of apples visible on the video captured from one side of the trees is less than the whole number of fruits present on the trees, and is higher than the number of fruits counted

on that side of the trellis—i.e. because trees are espaliered, so there are fruits hanging on the other side of the tree that are visible. Therefore, we had to manually annotate all the fruits on the videos and track them “by hand” to produce a reliable ground truth for the number of fruits that are actually visible in the videos. In the following section, that is the number we used as ground truth.

The datasets are summarised in Table 1. These correspond to the same datasets published by [8], which include erratic camera movement, low frame rates and intermittent occlusions of fruits. Therefore, the results presented here reflect the behaviour of the proposed system under real usage scenarios with those challenging conditions.

Table 1. Datasets description.

Dataset	# trees	# apples (ground truth)
Galafab-west	3	233
Schnico-Red-east	5	356
Schniga-Schnico-west	6	373

We evaluate the tracking and counting capacity of our system using the YOLOv4 model trained to detect mature apples (Sect. 2.2). A visual output of the fruit tracking algorithm is illustrated in Fig. 2.

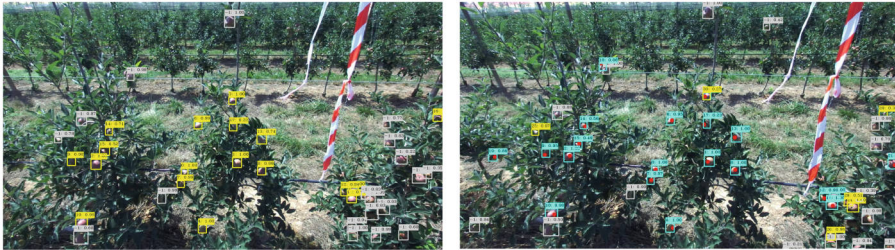


Fig. 2. The fruit tracking process. On the left image, which is frame 1, all the fruits are detected for the first time (yellow boxes). On the right image, which is frame 4, most of the fruits are the same of the previous frames, so they are represented in cyan boxes. The red dots are the projections of the 3D centroids of all fruits visible in previous frames onto the current frame. Grey boxes correspond to fruits that were detected by YOLO but whose estimated centroid lies outside of the detection box when re-projected onto the same frame, due to noise in the depth data, and hence are not considered in the current frame for fruit counting (Color figure online).

3.2 Results on Fruit Tracking and Counting

The results obtained with the proposed Algorithm (Sect. 2.3) and the algorithm proposed by [8] are shown on Table 2 and Table 3, respectively.

The Absolute Percentage Error (APE) metric shown on the Tables corresponds the quotient

$$APE = \frac{|Estimated - Actual|}{Actual} \times 100 \quad (2)$$

where *Estimated* is the number of fruits estimated by the algorithm and *Actual* is the actual number of fruits (ground truth).

The Multiple Object Tracking Accuracy (MOTA) is a metric used to evaluate object trackers that takes into account both the performance of the detection (i.e. false negatives (FN), false positives (FP)) and the number of identity switches (IDS) between tracked objects (i.e. tracking errors), and is computed as:

$$MOTA = 1 - \frac{FN + FP + IDS}{GT} \quad (3)$$

where *GT* is the ground truth number of objects to track.

Table 2. Fruit tracking and counting results with the algorithm we propose, which re-projects only fruit centroids.

Dataset	# apples (estimated)	# APE (%)	MOTA
Galafab-west	267	14.59	0.32
Schnico-Red-east	286	19.66	0.14
Schniga-Schnico-west	586	57.10	0.27

Table 3. Fruit tracking and counting results with the algorithm proposed in [8], which re-projects all the 3D points for each fruit.

Dataset	# apples (estimated)	# APE (%)	MOTA
Galafab-west	193	17.17	0.49
Schnico-Red-east	248	30.34	0.25
Schniga-Schnico-west	395	5.90	0.42

Overall, the results suggest that the system that we propose, either with algorithm proposed in this paper or the one proposed by [8], is able to successfully track fruits with high precision and provide acceptable errors for the total fruit count. Comparing the algorithm that we propose in this paper with that of

[8], we realise that re-projecting all the points, rather than only the centroids, makes the system more robust to errors generated by the object detector and the depth estimated by stereo. This conclusion stems from the higher MOTA values reported in Table 3, which indicate more stable tracks.

Our algorithm (Table 2) achieves slightly better results in terms of total fruit count on the first two datasets, but produces a much worse count in the third dataset due to over-counting, compared to the algorithm proposed by [8]. We believe this is due to noise in the camera extrinsics estimated in real-time, probably because the images in this dataset are more difficult to track from a visual odometry perspective. The increased robustness shown by the algorithm proposed in [8] comes from re-projecting all the points from each fruit and combining that with a Hungarian assignment. That approach reinforces the correct matches frame after frame, such that even if errors or noise occur later, the mass of points previously matched avoids a wrong match of new detections to fruits. On the other hand, when we project only the fruit centroids, since the tracking is less robust, the algorithm tends to overestimate the number of fruits, because it tends to lose track of a fruit more easily and thus a new tracking “object” is created for the same fruit, which ends up being counted several times.

The advantage of working with centroids is a less computationally demanding algorithm (less points to process), which makes it more suitable for real time usage on less powerful IoT devices, at the cost of a decreased accuracy. Moreover, the usage of a stereo camera instead of COLMAP (the SfM component used in [8]) allowed us to reduce the total computational time drastically. Overall, with the solution here proposed, we could reduce the total processing time for each dataset from 2–3 h to 1.5–2.5 min.

4 Conclusions

This work’s strong suit is its robustness when tracking fruits with intermittent occlusions (as in [8]) while keeping its applicability to devices with lower computational resources. Compared to the algorithm proposed in [8], the approach that we propose, based on 3D centroids, is computationally less demanding and thus more suitable to online settings, e.g. where a system with a stereo camera is used to obtain fruit counts in real time. For use cases where accuracy is more important than speed, our system can still be paired with the more robust algorithm proposed in [8] either with a stereo camera, such as the ZED 2i used in this work, or even with a SfM pipeline as described in [8].

The results shown in this paper were obtained with a ZED camera operated by hand, and the video processed offline. In the near future, we want to test our algorithm in an *online* setting, where the camera is attached to an agricultural vehicle and the depth maps for each frame are fetched in real-time from the ZED camera, as this would be the most realistic and scalable setup, capable of easily covering large field portions.

Besides the promising results shown here, we must look for the opportunities that a system like this can add to the precision fruit farming, compared

to the information obtained by the manual counting of a few trees, currently applied in modern fruit farms. In fact, even with errors, the overview of the entire orchard, resulting from the installation of the camera in a tractor makes real-time mapping possible, supporting technical decisions with major economical impact, such as supplementary fruit thinning, water or nutrient adjustments or harvest management. Historical data inside each orchard also allows different approaches in pruning strategies, adjusting the number of flower buds, as well as in the definition of an optimised fruit setting strategy. Combined with other solutions provided by remote sensing, it can be also a powerful tool for the application of variable rate technologies, such as precision fertilisation.

The algorithmic approach presented in this work, along with its future improvements, was done during the development of a software solution which is currently being built by SISCOG¹. This solution seeks to integrate and process data from different sensors (remote or not), and display the results in a user-friendly manner in order to support farmer's decision making. We believe that, by presenting the results of this research to farmers in this way, we are able to obtain valuable feedback that will allow us to define future directions for this work.

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